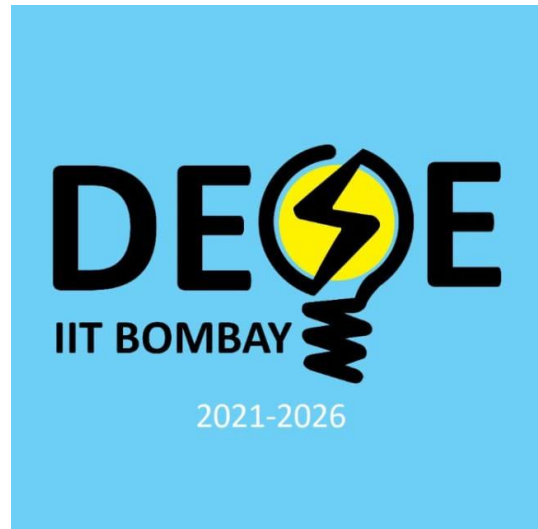


# Solar cooker Report



## Group 10

- Aditya Vibhute - 21D10003
- Rahul B - 21D170033
- Jayank Chaudhary - 21D170020

Under the guidance of

Prof. Shireesh Balwant Kedare

# Plan of Action

## Why box type?

- Box type cooker works with a simple principle and it can be made with things we use in our everyday life
- Easy for calculation.

## Apparatus

- **Box** - This forms the body of the solar cooker which is insulated from the inside. We used a cardboard box.
- **Aluminium foil / Mirror** – We used aluminium foil or mirror as a reflector to concentrate sunlight from sun inside the body. This reflector can be inclined at any angle to reflect the sunlight inside the box.
- **Glass / plastic cover** – We used glass or plastic as lid for the cooker and to trap the heat inside of the box.
- **Black paper/ cloth** – Black paper or cloth was used to insulate the box as it is a good heat absorber.
- **Thermocol** – Thermocol was used to insulate the interior of the box.
- **Cardboard** – We used cardboard as a support for the reflector.



## **Methodology**

- We used a typical cardboard box and insulated its walls and base with thermocol.
- We used cardboard as a reflector and covered it with aluminium foil. We also covered the sidewalls of the box with aluminium foil for good reflection.
- Then we covered the base with black paper or cloth to boost its heat absorption and for rapid cooking.
- Then we used plastic cover or glass as a lid for the cooker.
- Then we attached the reflector to the box to reflect all the sunlight inside the box.

## **Working Principle**

- This is a slow cooking process powered by sunlight
- Radiation enters the body of the cooker directly from sun and reflected by reflector through transparent lid
- The radiation inside is reflected on the sides and absorbed on the base so that the area near the base is hot where the food is placed
- Radiation does not leave the body as it is trapped by lid and it helps heat cook the food

# **Performance Assessment**

## **Design parameters**

### **Aditya's cooker**

- Cardboard box dimension = 37.5cm x 26.5cm x 31cm
- Dimensions of the reflector = 81cm x 38.5cm
- Aperture area:  $A_{ap, effective} = 39\text{cm} \times 38.5\text{cm} = 0.15\text{m}^2$
- Pot area:  $A_{pot} = 0.02 \text{ m}^2$
- Dimension of plastic lid = 37.5cm x 26.5cm
- Dimension of glass lid = 37.5cm x 26.5cm
- Projected absorber area:  $A_{abs} = 0.1 \text{ m}^2$
- Effective dimensions of the box (inner cavity to hold the utensil) = 37.5cm x 26.5cm x 14cm
- Optical concentration ratio:  $C_{optical} = A_{ap, effective}/A_{abs} = 1.5$
- Geometric or thermal concentration ratio:  $C = A_{ap, effective}/A_{pot} = 7.5$
- Steady state temperature of 75 ° C was achieved after 45 min.

### **Rahul's cooker**

- Cardboard box dimension = 23cm x 20.5cm x 10cm
- Dimensions of the reflector = 22.5cm x 20cm
- Aperture area:  $A_{ap, effective} = 34.9\text{cm} \times 23\text{cm} = 802.7 \text{ cm}^2 = 0.08 \text{ m}^2$
- Pot area:  $A_{pot} = 0.013 \text{ m}^2$
- Dimension of plastic lid = 23.5cm x 20.5cm
- Dimension of glass lid = 23.5cm x 20.5cm
- Projected absorber area:  $A_{abs} = 404.8\text{cm}^2 = 0.04 \text{ m}^2$
- Effective dimensions of the box (inner cavity to hold the utensil) = 20cm x 17.5cm x 8cm
- Optical concentration ratio:  $C_{optical} = A_{ap, effective}/A_{abs} = 1.98$
- Geometric or thermal concentration ratio:  $C = A_{ap, effective}/A_{pot} = 6.15$
- Steady state temperature of 80 ° C was achieved after 25 min.

## **Jayank's Cooker**

- Cardboard box dimension = 24cm x 25cm x 22cm
- Dimensions of the reflector = 20.5cm x 25cm
- Aperture area:  $A_{ap, effective} = 31\text{cm} \times 24\text{cm} = 744\text{cm}^2 = 0.075 \text{ m}^2$
- Pot area:  $A_{pot} = 0.011 \text{ m}^2$
- Dimension of plastic lid = 24cm x 25cm
- Dimension of glass lid = 24cm x 25cm
- Projected absorber area:  $A_{abs} = 441.2\text{cm}^2 = 0.044 \text{ m}^2$
- Effective dimensions of the box (inner cavity to hold the utensil) = 21cm x 20cm x 20cm
- Optical concentration ratio:  $C_{optical} = A_{ap, effective}/A_{abs} = 1.98$
- Geometric or thermal concentration ratio:  $C = A_{ap, effective}/A_{pot} = 6.81$
- Steady state temperature of 65 ° C was achieved after 30 minutes.

## **Operating parameters**

### **Aditya's cooker**

- Surrounding temperature = 34°C
- Solar radiation falling on Earth = 800 W/m<sup>2</sup>

### **Rahul's cooker**

- Surrounding temperature = 35°C
- Solar radiation falling on Earth = 800 W/m<sup>2</sup>

### **Jayank's cooker**

- Surrounding temperature = 30°C
- Solar radiation falling on Earth = 800 W/m<sup>2</sup>

## **Performance parameters**

First figure of merit is a factor we measure to compare the performance of various cookers. We keep water in cooker and wait till it reaches steady state. Then we can calculate F1 as shown below.

➤ First figure of merit  $F_1 = (T_{\text{pot}} - T_{\text{amb}}) / (C \times G_T)$

- $T_{\text{pot}}$  = Temperature of pot
- $T_{\text{amb}}$  = Ambient temperature outside
- $C$  = Thermal concentration ratio
- $G_T$  = Solar radiation

### **Aditya's cooker**

Aditya's calculation for First Figure of Merit

$$F_1 = (75-34) / (7.5 \times 800) = 0.0068 \text{K-m}^2/\text{W}$$

### **Rahul's cooker**

Rahul's calculation for First Figure of Merit

$$F_1 = (80-35) / (6.15 \times 800) = 0.0091 \text{K-m}^2/\text{W}$$

### **Jayank's cooker**

Jayank's calculation for First Figure of Merit

$$F_1 = (65-30) / (6.81 \times 800) = 0.0064 \text{K-m}^2/\text{W}$$

## **Questions**

### **❖ What can be cooked using solar cooking?**

We can use this type of solar cooker to cook food that requires a constant temperature of 70° C to 80° C. For example, rice, fatless omelet, vegetables, teas, noodles, etc.

### **❖ What cannot be cooked using solar cooking?**

We cannot cook food that requires oil or ghee since the temperature needed to heat oil or ghee is over 100 ° C. For example, several types of beans, idli, dal, etc.

### **❖ Does food cooked with solar cookers taste different (better/worse) than food cooked with other fuels?**

There was no difference in the taste of rice, the fatless omelette tasted better than the omelette that fried in pan. Food cooked using solar cookers are proven to be more nutritious because of controlled heat, whereas in gas stove due to excess rapid heat many nutrients are lost.

### **❖ How do different solar cookers compare?**

Solar cookers can be compared on various parameters such as: First figure of merit, efficiency, life span, speed, capacity, cost, maintenance, weight, etc.

### **❖ How would you compare solar cooking to LPG cooking? Specify performance parameters.**

Solar cookers are slow and can only cook certain foods whereas LPG can cook fast. But solar cookers are environment friendly and LPG harms the environment and consumes lot of resources. So, each type of energy source has its advantages and disadvantages. Few performance parameters are Thermal efficiency, Figure/s of merit, Time constant

### **❖ Determine the efficiency of solar cooking**

Efficiency of the solar cooker by defined the ratio of the energy utilized to cook the food and the energy supplied to the cooker.

## ❖ Can we use this for a community kitchen (about 200 inmates)?

We cannot use box solar cooker for such a large kitchen since it can only achieve a max temperature of up to  $100^{\circ}\text{C}$  and it also has low capacity. But we certainly can use different solar cookers such as parabolic solar cookers, etc. which can achieve remarkably hot temperatures in short period of time.

## Experience

### **Aditya's experience**

This was my first group project in IITB which I would have preferred to do on the campus with my group members but unfortunately, we had to do it at our home. Doing this project taught me the patience that is needed to perform an experiment, it taught me how to work with thermocol, it also taught me how to cook rice (since I never cooked rice before). I had a very pleasant experience working with my group members and interacting with the professor during the presentation that was conducted on solar cookers.

### **Rahul's experience**

This was quite a pleasant experience considering the situation. And this is the better way of learning things, by experiments. It helped me make a transition from being a part of the problem statement to trying to solve the problem as an engineer. We had a fun time as a team. We had a wonderful presentation organized by our professor, in which I saw and learnt how far we can take even a simpler project like solar cooker in order to make it more effective and more user friendly.

### **Jayank's experience**

For me this experiment was one of the things that reminds me that I am in college now. By doing this experiment I'm able to see the difference in theoretical and practical knowledge, like theory cannot



teach us everything, there is some problems which can only be seen by practically doing things. My parents were also curious about solar cooker project like how I am going to make a cardboard box into solar cooker. This was the first time I am dealing problems like an engineer. It was my second group project and I had fun in doing this specially during the presentation when professor was cross questioning. Overall it was a great experience for me.

## **Annexure**

### **Rice by Aditya**

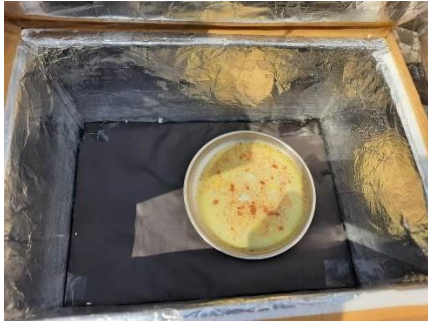
#### **Before**



#### **After**



**Omelet by Aditya**  
**Before**



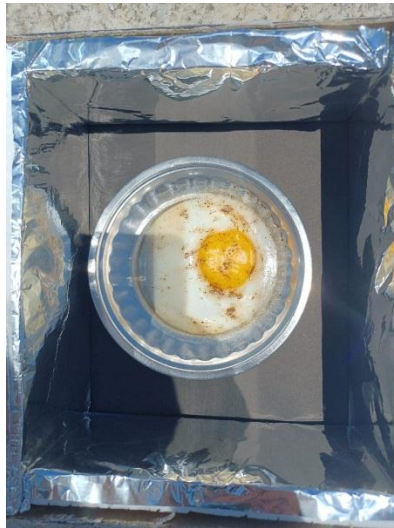
**After**



**Omelet by Rahul**  
**Before**



**After**



## Rice by Jayank

### Before



### After



## COOKERS



From left to right (Aditya, Rahul, Jayank)

\*\*\*\*\*